

Creating an Azure Cosmos DB Account using the Table API

Summary:-

In this lab, you will create an Azure Cosmos DB account using the **Table API**. You will create a table, add an entity, and view the entity properties in the Data Explorer. This lab demonstrates how the Azure Cosmos DB Table API stores key-value data similar to Azure Table Storage.

Let's Start:-

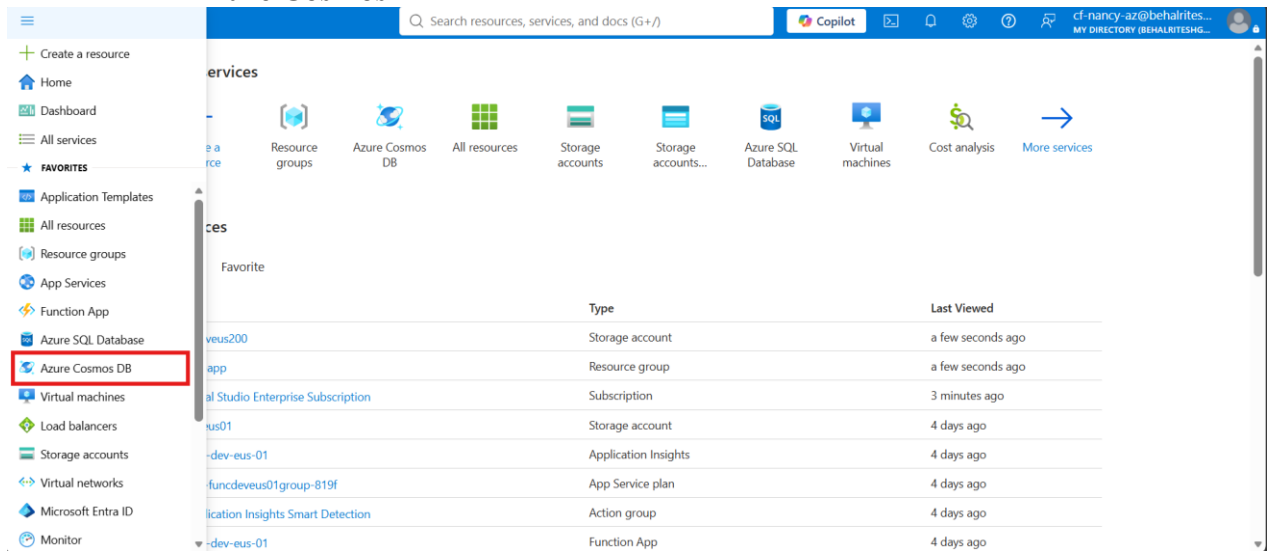
After completing this lab, you will be able to:

- Create an Azure Cosmos DB account using the Table API.
- Create a table in Azure Cosmos DB.
- Add entities to a table.
- View entity properties including system-generated timestamps.
- Explore data using Data Explorer.

Lab Steps:-

Step 1 – Select the Table API

- Sign in to the **Azure Portal**.
- Select **Azure Cosmos DB**.



- Click **Create**.

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Home

Azure Cosmos DB ★ ...

Identify non-compliant Azure Cosmos DB accounts using ARG. Filter Azure Cosmos DB accounts by resource group. How many Azure Cosmos DB accounts do I have?

+ Create Restore Manage view Refresh Export to CSV Open query Assign tags Add to service group Group by none

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

No Azure Cosmos DB accounts to display

Create a globally distributed, multi-model, fully managed database using API of your choice. Or try it for free, up to 20k RU/s, for 30 days with unlimited renewal.

+ Create

Try now

Showing 1 - 0 of 0. Display count: auto

Give feedback

- Under **Other APIs**, select **Azure Cosmos DB for Table**.
- Click **Create**.

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Home > Azure Cosmos DB

Create an Azure Cosmos DB account ...

Which API best suits your workload?

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. [Learn more](#)

To start, select an API to create a new account. The API selection cannot be changed after account creation. To use multiple APIs, you can create separate accounts with different APIs as needed.

Recommended APIs **Other APIs**

Azure Cosmos DB for Table
Fully managed database service for apps written for Azure Table storage. Recommended if you have existing Azure Table storage workloads that you plan to migrate to Azure Cosmos DB.
Create [Learn more](#)

Azure Cosmos DB for Apache Gremlin
Fully managed graph database service using the Gremlin query language, based on Apache TinkerPop project. Recommended for new workloads that need to store relationships between data.
Create [Learn more](#)

Azure Cosmos DB for Apache Cassandra
Fully managed Cassandra database service for apps written for Apache Cassandra. Recommended if you have existing Cassandra workloads that you plan to migrate to Azure Cosmos DB.
Create [Learn more](#)

Azure Cosmos DB for MongoDB RU
Fully managed MongoDB-compatible database service using the request unit (RU) provisioning model. Recommended if you have existing RU-based MongoDB workloads, or need point-reads, simple queries, and granular RU autoscaling.

Azure Cosmos DB for PostgreSQL
Fully-managed relational database service for PostgreSQL with distributed query execution, powered by the Citus open source extension. Build new apps on single or multi-node clusters—with support for JSONB, geospatial, rich indexing.

- This step starts the creation of an Azure Cosmos DB account using the Table API.

Step 2 – Configure the Cosmos DB Account

- Configure the following settings:
 - **Workload Type:** Learning
 - **Subscription:** Select your Azure subscription.
 - **Resource Group:** Select an existing resource group or create a new one.
 - **Account Name:** Enter a unique name.
 - **Availability zone:** Disable

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Home > Azure Cosmos DB > Create an Azure Cosmos DB account

Create Azure Cosmos DB Account - Azure Cosmos DB for Table

Project Details

Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.

Workload Type *

Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group *

[Create new](#)

Instance Details

Account Name *

Configure availability zone settings for your account. You cannot change these settings once the account is created.

Availability Zones ☐ Enable ☒ Disable

[Review + create](#) [Previous](#) [Next: Global distribution](#) [Feedback](#)

- **Region:** East US / East US 2.
- **Capacity Mode:** Provisioned Throughput.
- **Apply Free Tier Discount:** Apply
- Keep the **Limit Total Account Throughput** enabled.
- Click **Next**.

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Create Azure Cosmos DB Account - Azure Cosmos DB for Table

Location *

Available locations are determined by your subscription's access and availability zone support (if that is enabled). If you don't see or cannot select your desired location, please open a support request for region access.
[Click here for more details on how to create a region access request](#)

Capacity mode

☒ Provisioned throughput

Request units per second (RU/s) are pre-configured and billed hourly, offering guaranteed throughput with two modes—**Autoscale**, which dynamically adjusts within a 1-10x range for varying workloads, and **Manual**, with fixed RU/s for steady, predictable traffic.

☐ Serverless

Ideal for intermittent or unpredictable traffic. Billed only for consumed RU/s, scaling on-demand without capacity planning or minimum charges.
[Learn more about capacity mode](#)

With Azure Cosmos DB free tier, you will get the first 1000 RU/s and 25 GB of storage for free in an account. You can enable free tier on up to one account per subscription. Estimated \$64/month discount per account.

Apply Free Tier Discount ☒ Apply ☐ Do Not Apply

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- This step configures the Azure Cosmos DB Table API account.

Step 3 – Configure Deployment Settings

- On the **Global Distribution** tab, leave the default settings unchanged.

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Create Azure Cosmos DB Account - Azure Cosmos DB for Table

Basics **Global distribution** Networking Backup Policy Security Tags Review + create

Configure global distribution and regional settings for your account. You can also change these settings after the account is created.

Geo-Redundancy ☒ Enable ☐ Disable

Multi-region Writes ☐ Enable ☒ Disable

Review + create Previous **Next: Networking**

Feedback

- Click **Next**.
- On the **Networking** tab, select **Enable access from all networks**.

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Basics Global distribution **Networking** Backup Policy Security Tags Review + create

Network connectivity

You can connect to your Azure Cosmos DB account either publicly, via public IP addresses or service endpoints, or privately, using a private endpoint.

Connectivity method *

☒ **All networks**
☐ Public endpoint (selected networks)
☐ Private endpoint

All networks will be able to access this CosmosDB account. <http://aka.ms/network-security>

Connection Security Settings

Minimum Transport Layer Security Protocol

Review + create Previous **Next: Backup Policy**

Feedback

- Leave the remaining settings as their default values.
- In the **Backup Policy** tab, select **Periodic** as the Backup policy. Leave the **Backup interval** and **Backup retention** settings at their default values.

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Basics Global distribution Networking **Backup Policy** Security Tags Review + create

Azure Cosmos DB provides three different backup policies. You will not be able to switch to Periodic mode once you adopt Continuous mode. [Learn more](#) about the differences of the backup policies and pricing details.

Backup policy ⓘ

☒ Periodic
Backup is taken at periodic interval based on your configuration

☐ Continuous (7 days)
Provides backup window of 7 days / 168 hours and you can restore to any point of time within the window. This mode is available for free.

☐ Continuous (30 days)
Provides backup window of 30 days / 720 hours and you can restore to any point of time within the window. This mode has cost impact.

☐ Continuous (35 days)
Provides backup window of 35 days / 840 hours and you can restore to any point of time within the window. This mode has cost impact.

Backup interval ⓘ

240 Minute(s)
60-1440

Backup retention ⓘ

8 Hours(s)
8-720

Review + create Previous **Next: Security** Feedback

- Click Next.
- On the **Security tab**, leave the default settings unchanged. Click Next.

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Basics Global distribution Networking Backup Policy **Security** Tags Review + create

Key-based Authentication

Key-based authentication uses account keys that are granted permissions to the entire Cosmos DB account. Disabling these methods improves security by ensuring that Cosmos DB database accounts exclusively require Entra ID identities for authentication. [Learn More](#)

Key-based Authentication ⓘ ☒ Enable ☐ Disable

Data Encryption

Azure Cosmos DB encryption protects your data at rest by seamlessly encrypting your data as it's written in our datacenters, and automatically decrypting it for you as you access it.

By default your Azure Cosmos DB account is encrypted at rest using service-managed keys. At the moment, you will not be able to switch back to service-managed key after opting into using custom-managed key while creating your account. [Learn More](#)

Data Encryption * ☒ Service-managed key ☐ Customer-managed key (CMK)

Review + create Previous **Next: Tags** Feedback

- On the **Tags tab**, leave the default settings unchanged.

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Basics Global distribution Networking Backup Policy Security **Tags** Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [learn more](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

☐ Key	Value	Resource Type
		Azure Cosmos DB account

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- Click **Next: Review + create**.
- On the **Review + Create** tab, verify the configured settings. After the validation succeeds, **click Create**.

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Create Azure Cosmos DB Account - Azure Cosmos DB for Table

Basics Global distribution Networking Backup Policy Security Tags **Review + create**

✓ Validation Success

Creation Time

Estimated Account Creation Time (in minutes) 3

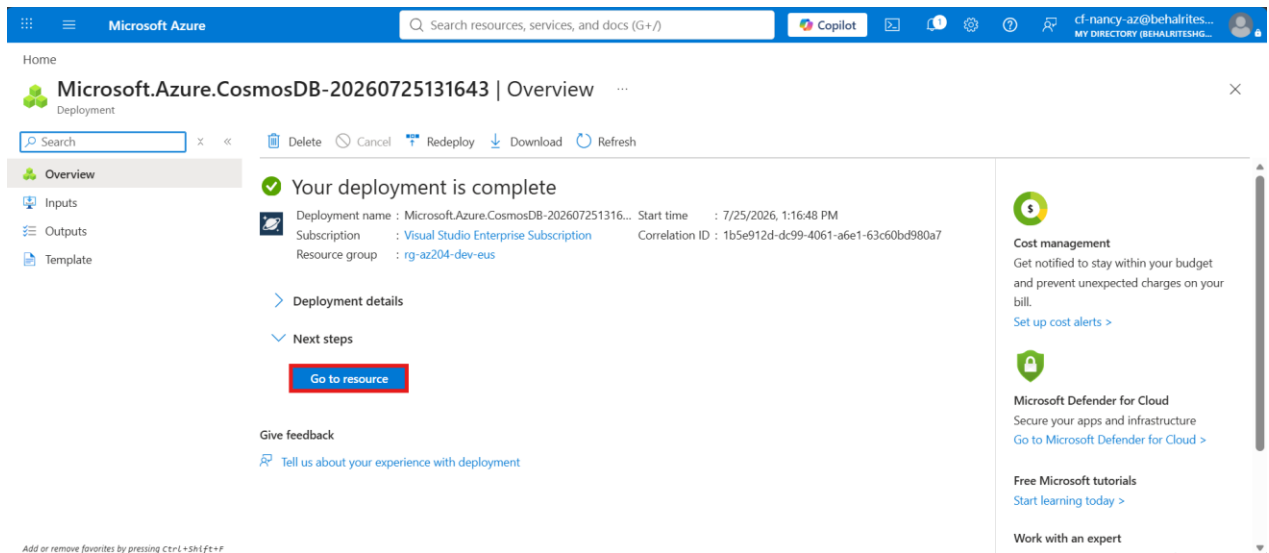
The estimated creation time is calculated based on the location you have selected

Basics

Subscription	Visual Studio Enterprise Subscription
Resource Group	(new) rg-az204-dev-eus
Location	East US 2
Account Name	(new) app-cosmos-db-2
API	Azure Cosmos DB for Table
Capacity mode	Provisioned throughput

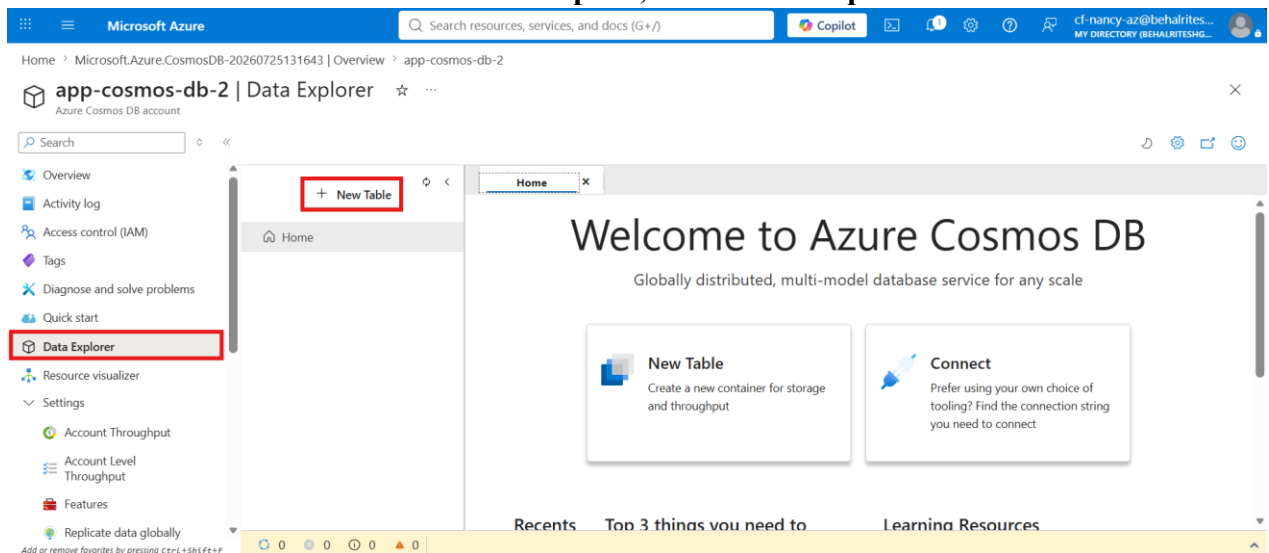
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- Wait for the deployment to complete successfully. Then, it will appear as:

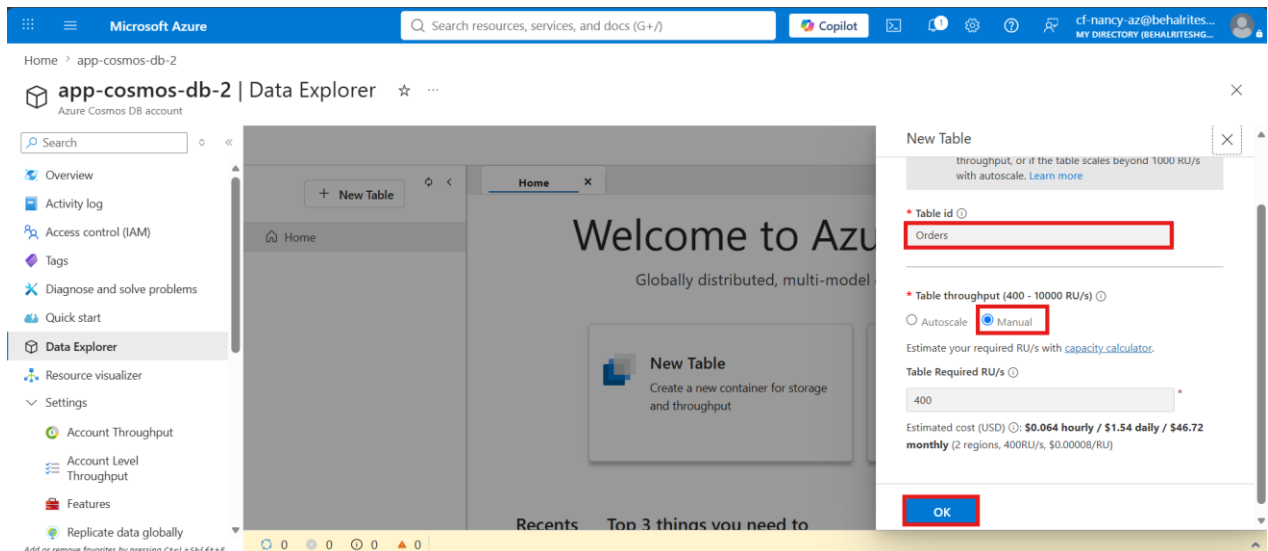


Step 4 – Create a Table

- Click **Go to resource**. From the left pane, select **Data Explorer**.



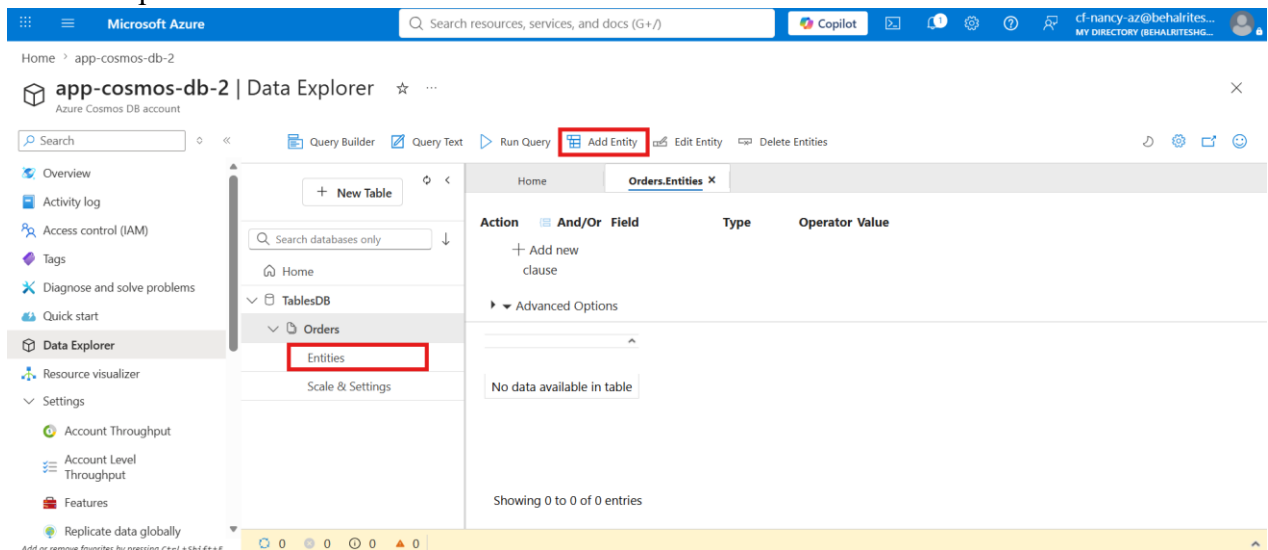
- Click **New Table** to create a table. **Enter** the Table id as Orders.
- Select **Manual** for Table throughput. **Set Table** Required RU/s to 400.



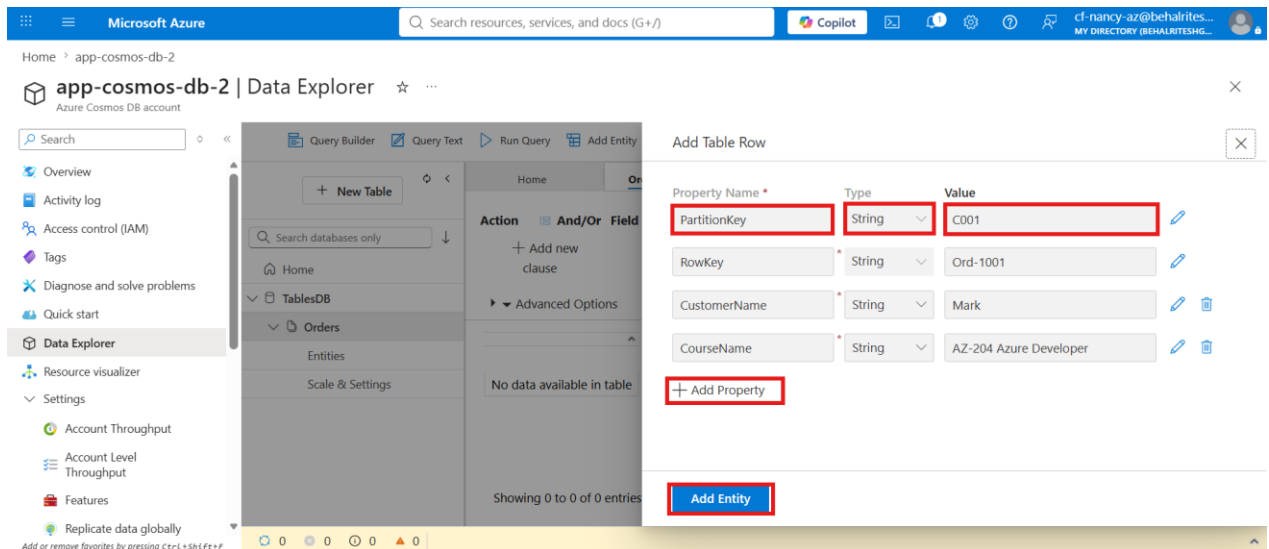
- Click **OK**. This step creates a table in the Azure Cosmos DB Table API account using manual throughput.

Step 5 – Add an Entity

- Expand the **Orders** table and select **Entities**.

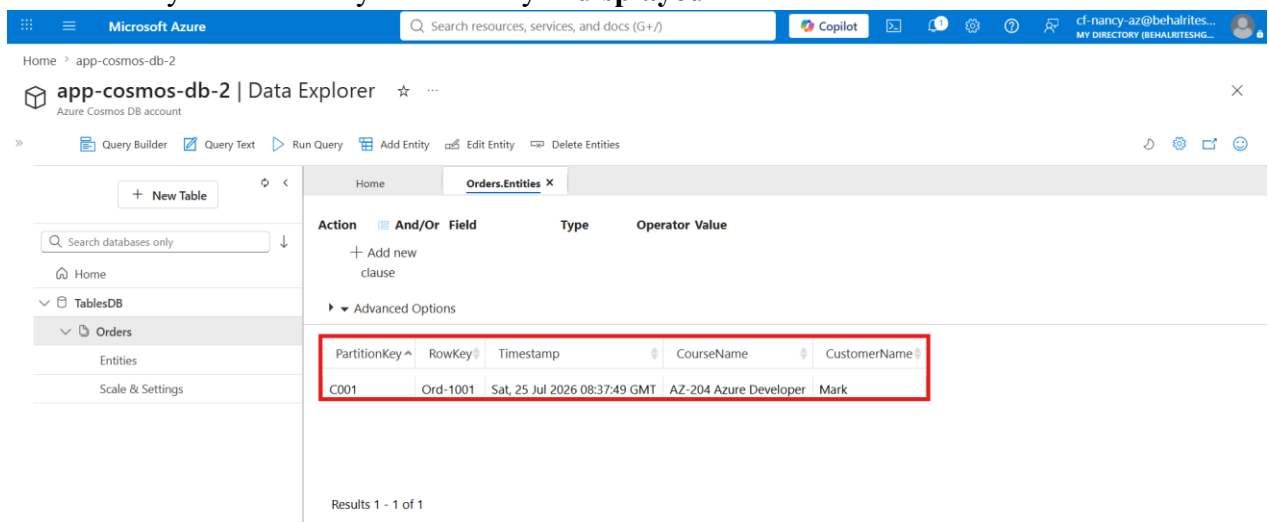


- Click **Add Entity**.
- Enter the following values:
 - **Partition Key:** C001
 - **Row Key:** ORD-1001
- Add the following properties and set their Type:
 - **CustomerName** – String – Mark
 - **CourseName** – String – AZ-204 Azure Developer
- Similarly, you can add more properties and select the required data type. Click **Add Entity**.



Step 6 – View the Entity

- Select the **Orders** table and open **Entities**.
- Verify that the newly added entity is **displayed**.



- **Observe** the automatically generated Timestamp property.
- This step verifies that the entity is stored successfully and that Azure Cosmos DB automatically generates system properties.

Verification

Verify the following:

- Azure Cosmos DB Table API account is created successfully.
- The **Orders** table is created.
- An entity is successfully inserted into the table.
- The entity displays the configured properties.
- The **Timestamp** property is automatically generated by Azure Cosmos DB.

